

EE 4113 – ECE Lab 2
Fall 20XX
Lab #XX Report

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Objective:

[Briefly describe the educational and conceptual objectives.]

Equipment List:

Table 1 Equipment List

<i>Description</i>	<i>Manufacturer</i>	<i>Model Number</i>
Pluggable Breadboard	R.S.R. Electronics	MB-108
Power Supply	Lab-Volt	AA771
Digital Bench Multimeter	Tektronics	DM2510

Parts List:

Table 2 Parts List

<i>Quantity</i>	<i>Part</i>	<i>Value</i>
1	Resistor	1.2k Ω , 1W
1	Resistor	1.8k Ω , 1W
1	Resistor	2.2k Ω , 1W
1	Resistor	3.3k Ω , 1W
1	Resistor	4.7k Ω , 1W
1	Resistor	5.6k Ω , 1W

Procedure:

Part A: the procedure was setting the current limit using the probes.

[Briefly describe the steps/procedure performed.]

Part B:

Part C:

Results:

Part A: 1.7 V minimum (low) 2.14 V Brighter

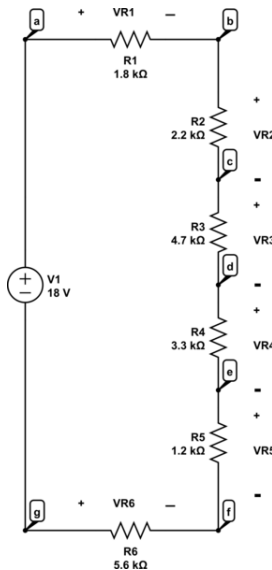
[Place only information about the direct measurements/results.]

Part B: 15.3 V @ 18 mA

Part C:

$$P = (14.9 \text{ mA})^2 (668) = 149 \text{ mW}$$

$$I_{LED} = \frac{9.4\%}{668} = 14.9 \text{ mA}$$



Part D: $V_R = 13V$

$$R = \frac{13V}{3(26 \text{ mA})} = 216.67 \Omega \approx 220 \Omega$$

Largest available = 220 Ω

$$P = (3 \cdot 20 \text{ mA})^2 (216.67) = 0.780 \rightarrow 780 \text{ mW}$$

$$I_{LED} = (3 \cdot 20 \text{ mA}) = 60 \text{ mA}$$

$$V_{1LED} = 1.9V$$

$$V_{3LED} = 1.9V$$

$$V_{2LED} = 1.9V$$

Figure 1: Example, figure captions go on bottom.

Table 3: Example, table captions go on top.

Node	Measured Voltage (V)
$V_{R1} = V_{ab}$	1.7041V
$V_{R2} = V_{bc}$	2.1064
$V_{R3} = V_{cd}$	4.5

$V_{R4} = V_{de}$	3.1596
$V_{R5} = V_{ef}$	1.1489
$V_{R6} = V_{fg}$	5.3617

Conclusion:

[Describe the meaning behind the measurements/results. You may share your perspective on the quality of the performance.]